Adel Bazzi

Final Project Proposal

Project Title:

The title of my project is “Cartoon Animal Classification”.

Student Group:

I will be working on this project by myself. I will be in charge of data acquisition, data processing, model development, model training, and evaluation.

Abstract:

This project involves training a machine learning model to recognize and classify different cartoon animals from images. The goal is to develop a model that can accurately identify animal categories based on visual features such as shape, color, facial features, and body structure. Although the architecture has not yet been finalized, a convolutional neural network (CNN) is expected to be a suitable choice due to its ability to extract spatial features from images and recognize visual patterns. The project will involve collecting and preparing a dataset of cartoon animal images, training the model, and evaluating its performance using appropriate classification metrics. Different approaches to data preprocessing, model architecture, and hyperparameter tuning may be explored to improve accuracy and generalization. Ultimately, this project aims to demonstrate the effectiveness of deep learning techniques for image classification in stylized artwork, while examining the challenges associated with distinguishing between cartoon animals that may vary significantly in appearance, artistic style, and level of detail.

Timeline:

1. Week 1- Acquire dataset from Kaggle
2. Week 2- Complete data pre-processing
3. Week 3,4- Develop model architecture
4. Week 5- Train model
5. Week 6- Evaluate model
6. Week 7- Hyperparameter tuning